

1 Few Essential Rules

- The assignments should be completed in groups of two students.
- At least one of the team members must submit the assignment to Moodle.
- Discussions are allowed and encouraged. While working in groups, try using a version control system like GitHub.

2 Assignment Statement

Draw five (or ten) equispaced lines (assume 1 unit distance between any two lines) parallel to x , y , z -axis. So, all these lines together would create a 3D cubic grid structure of size $5 \times 5 \times 5$ or $10 \times 10 \times 10$ cubic units.

Draw a solid single-colored cube of size $1 \times 1 \times 1$ cubic units at some place inside the cubic grid. Ensure that the smaller cube's edges coincide with the grid lines. Use six keys (left, right, up, down, U, B) to move the cube along the three axes in steps of ± 1 unit. The smaller cube must not leave the cubic grid.

Use the key 'C' to change the color of the solid cube. Once the 'C' key is pressed, a prompt will appear in the terminal to read the user-defined RGB color values for the cube. The cube's color should be set to the user-defined color.

If the solid cube is in a particular position (say at (2,2,3)) and the key 'F' is pressed, the unit element of the cubic grid (the $1 \times 1 \times 1$ portion at (2,2,3)) will be filled with the same color as the solid cube. Similarly, if the user wants to clear an already-filled color unit, he/she should move the solid cube to that position and press 'W' key. Note that once a portion of the grid is colored, it will remain so until cleared.

Use four keys (L, R, T, D) to rotate the cubic grid model (along with its colored portions) in all four directions. Note that when the cubic grid is rotated, the smaller solid-color cube inside must rotate accordingly without changing its position or orientation.

3 Submission Details

Draw a simple model using your code, e.g., a 3D line/circle in a rasterized fashion. Make your model colorful and visually appealing (use your imagination). Upload your clean code repository (no binary files, no unnecessary libraries) to Moodle on or before **21st September 2026**.

Each submission must include a report detailing all team members, the working principle of your codebase, and screenshots of the output. Additionally, it is recommended that you briefly explain your code and implementation details in the report.

There may be some bonus marks if you do something more than the assignment asked/ make something compelling.